NEXT GEN
SOFTWARE HUB

CYBER SECURITY INTERNSHIP

WEEK 06
TASK 06 REPORT

Network Enumeration

Using Nmap, Enum4linux and Netdiscover for Information Gathering

TOPICS

- Host Discovery
- Service Detection
- Network Enumeration

TOOLS

- Nmap
- Enum4linux
- Netdiscover

RESOURCES

- TryHackMe: Network Services

TASK

Perform host discovery & service detection on a local VM.

Phase 2: Scanning & Penetration Testing (Week 5–10)

PLANS :

- Week 6: Network Enumeration



PREPARED BY

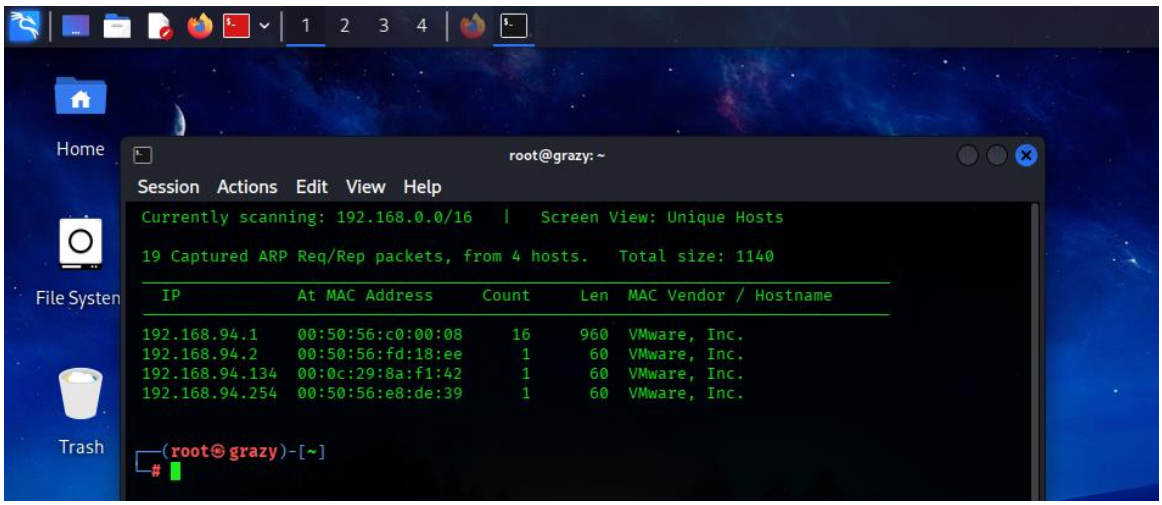
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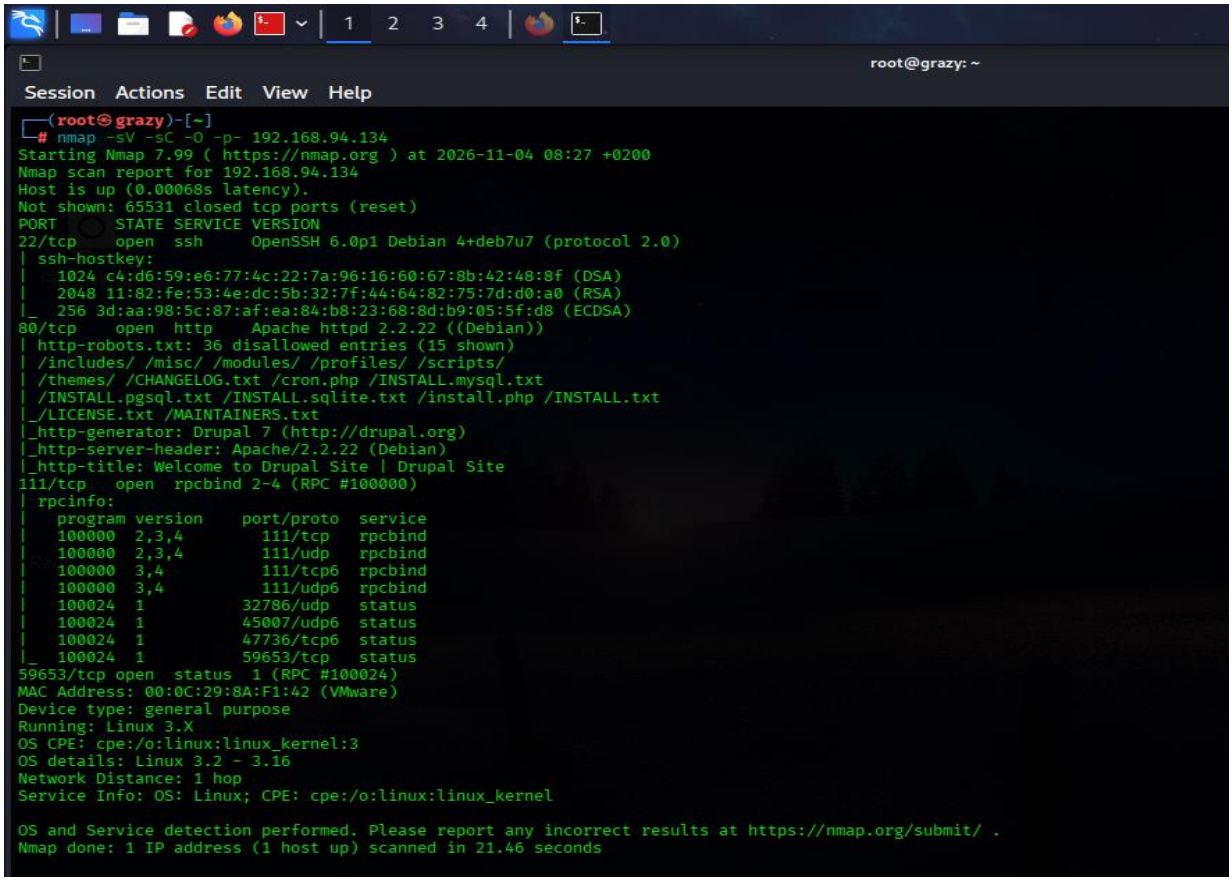
Introduction and Objective
The objective of this task was to move beyond basic port scanning and perform comprehensive network enumeration on a target virtual machine. While a standard scan simply tells us which network doors are open, enumeration is the process of looking through those doors to understand exactly what is inside. We needed to identify active hosts on our local network, determine the specific software and versions running on them, and attempt to extract additional system information to map out the target's attack surface.

Phase One: Host Discovery and Initial Mapping
We began by utilizing Netdiscover to map out the local virtual network environment. This tool relies on Address Resolution Protocol requests to identify live machines communicating on the subnet.



The results immediately highlighted our target machine, DC-1, operating at the IP address 192.168.94.134. By examining the MAC address vendor prefix, we could confirm that this was a VMware virtual machine, which aligns perfectly with our virtual lab setup. Following this discovery, we performed a lightweight Nmap ping scan to verify that the host was online and actively responding to network traffic.

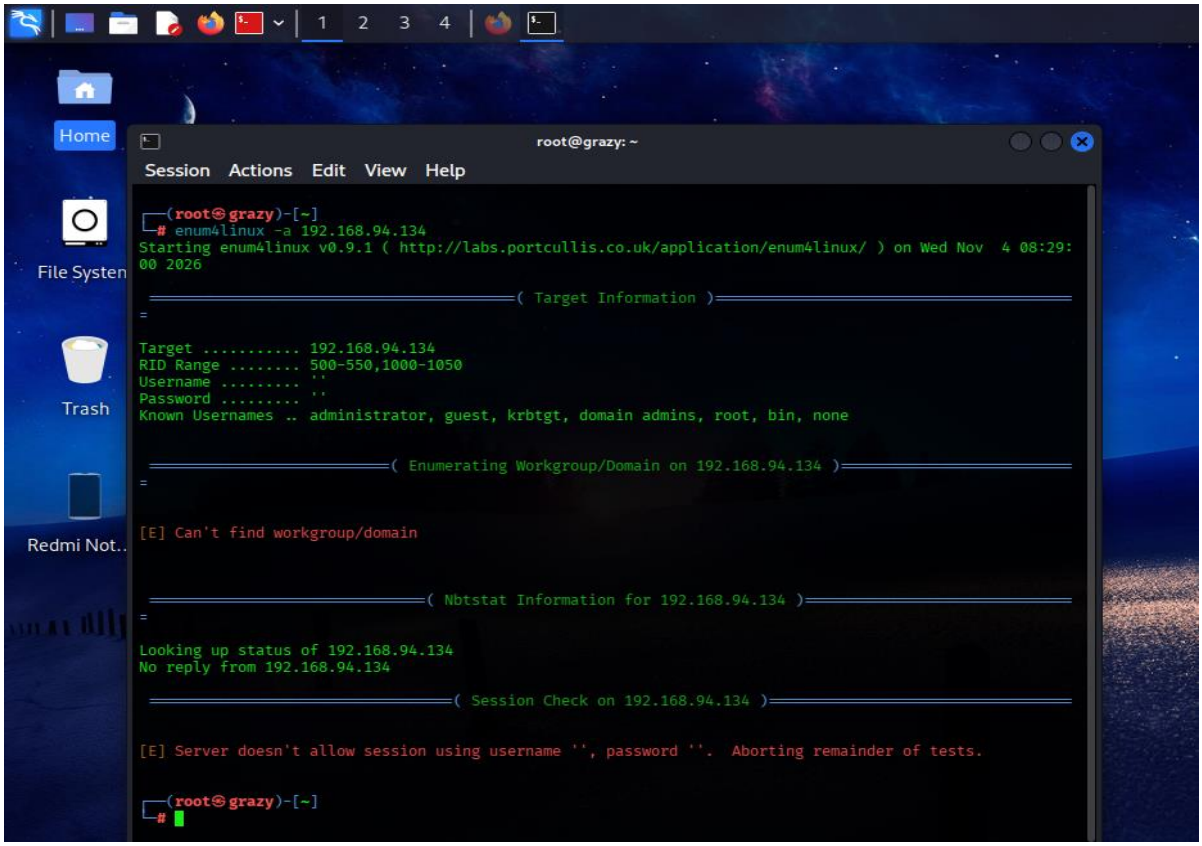
Phase Two: Service Detection and Script Scanning
With the target confirmed as active, we proceeded to the most critical part of the enumeration phase. We executed a comprehensive Nmap scan configured to check all sixty five thousand five hundred and thirty five TCP ports. Furthermore, we enabled service version detection and default script scanning to pull as much metadata as possible from the target.



The scan results provided a wealth of information. We identified three open ports. Port twenty two was running an SSH service, port eighty was running an Apache web server, and port one hundred eleven was running an RPCbind service. The most valuable intelligence, however, came from the Nmap scripting engine. By querying the web server on port eighty, Nmap extracted the page title, which clearly read "Welcome to Drupal Site". It also confirmed that the underlying Content Management System was Drupal version seven, and it successfully retrieved a list of disallowed entries from the robots text file. Discovering paths like the MySQL installation file and the main install PHP file is a massive indicator that the web application may have configuration weaknesses worth exploring later.

Phase Three: SMB Enumeration Attempt

After identifying the primary network services, we attempted to enumerate Server Message Block shares using Enum4linux. This tool is traditionally very powerful for extracting user lists, shared folders, and password policies from Windows environments.



In this specific scenario, however, the tool completely failed to gather any information. The output displayed several errors, stating that it could not find a workgroup or domain, and that the server did not allow the session. This was actually a very valuable learning experience. The enumeration failed because our target, DC-1, is a Linux based web server. Our initial Nmap scan had already proven that standard Windows networking ports, specifically ports one hundred thirty nine and four hundred forty five, were completely closed. Because Enum4linux relies entirely on those specific ports to communicate, it had no way to interact with the target. This highlighted a crucial rule in cybersecurity: tools must be strategically chosen based on the specific environment and the open ports discovered during the initial scanning phase.

Conclusion

The network enumeration phase was highly successful in mapping the target's attack surface. We established that DC-1 is a Linux system heavily focused on web services, specifically an outdated Drupal seven installation. The failed Enum4linux attempt was just as informative as the successful Nmap scan, proving that a security analyst must understand underlying network protocols to choose the right tool for the job. The intelligence gathered here, particularly the web server version and the exposed installation files, provides a clear direction for the next phase of vulnerability testing.